

Zeitgeist

A holographic interface for visualization of Flow using deep-learning models

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ABSTRACT

Zeitgeist is a participatory artwork that forms part of the AHRC-funded research project. The interface employs artificial intelligence (ai)-based algorithms to represent mental states of creative 'Flow': Flow, a concept of peak performance, low stress and heightened creative stimulation [Nakamura and Csikszentmihalyi, 2002] is measured and assessed using physiological data analysed in real-time using a deep-learning algorithms: Mapped onto 3D representations and displayed on a Pepper's ghost type hologram, participants jointly observe their individual and collective propensity of being in Flow. As a research project, the artefact is embedded into research on the effect of participatory art engagements on Flow, wellbeing and social connectedness. A pilot with two cohorts creatives (N=12) and scientists (N=16) demonstrates a significant effect of the participatory art intervention on Flow, mood and social connectedness, with facilitation and environmental context as confounding factors.

KEYWORDS

VR/AR, Machine Learning, Research, Art & Design

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1 INTRODUCTION

Zeitgeist is designed as a collaborative, participatory artwork based on physiological measures using EEG headsets to monitor if participants are in a Flow state - a creative state of mind increased creative stimulation, and heightened mood. Using a proprietary deep-learning algorithms derived from Flow state brainwave signature patterns, brain activity as measured by an electroencephalogram (EEG) is compared to these signature patterns, and resulting probability values are analysed for Flow states and then visually represented on a Pepper's ghost type hologram. Two Flow state probability values are mapped onto colour and shape values for two distinctive shapes. Lighter and more vibrant colours and complex shapes present heightened Flow states, darker colours and simpler forms symbolise lower likelihood of being in Flow. Flow-state cues

are destined to encourage Flow by providing incentives through visual neurofeedback - a process called 'nudging'. Based on Dr Rahman's research, which identified biomarkers of creative Flow states within creative improvisations [Rahman et al., 2021], such signature patterns of Flow are compared in real-time to ongoing participant interactions using AI. Trained and modelled on biomarker data, Rahman's deep-learning algorithm analyses and classifies incoming participant brainwave data from EEGs as real-time identifiers of Flow [Adnan et al., 2020].

The effect of participatory art on social connectedness has been evidenced [Gingrich et al., 2019], with higher social connectedness having been associated with better health and psychological wellbeing [Cohen, 2004], [Stansfeld and Candy, 2006]. The Zeitgeist artwork is using participatory art to advance 'Social Flow' - a Flow mental state based on social interaction. The research question explores if 'Social Flow' leads to a higher perceived social connection and sense of closeness, affects and mood. Here, two controlled pilot studies are presented, both conducted in collaboration with partners CNWL NHS Foundation Trust, Brunel University and the lead research institution University of Greenwich. The first intervention (Somerset House) involved six pairs of creatives (n=12), and the second (Royal Society) involved creatives and knowledge workers (n=16) with males and females between age 18 – 68. Both studies followed the same research protocol but differed in type of cohorts and environmental context. Analysis of pre- and post-intervention questionnaires, interviews and participant observations shed light onto key factors of social flow. Creative exercises encouraged joy, speed and a cognitive challenge as potentially contributing factors to Flow. Overall, users reported significantly deeper social connectedness, and sense of closeness post-intervention: Research objectives aim in recommendations on design and facilitation of online participatory art interfaces. Flow improves mood and happiness [Diener, 2000]. 'Flow' mental training is proposed to enhance cognitive performance, creativity and productivity. The relationship between Flow and social connectedness remains understudied. This pilot study addresses this.

2 ZEITGEIST INTERFACE

Incoming raw EEG data is analysed via NeuroCreate's proprietary deep-learning algorithm [Adnan et al., 2020] to calculate the continuous probability of Flow states in real-time. This probability measure is a floating points value between 0 and 1 which is then parsed using an Open Sound Control (OSC) protocol, and sent to a TouchDesigner (Derivative) patch. This allows for real-time representation of these incoming values, which are mapped onto two 3-dimensional Moebius strips (FBX files) - one for each participant in a given pair. The colour of the Moebius strips responds to the participants' probability of being in Flow, with warmer colours (golden

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and warm teal/blue tones) indicating a likely Flow state and cooler colours (purple/violet tones) the potential absence of Flow. The scale and visual complexity of the resulting shape are influenced by a shared Flow factor, computed by adding both participants' Flow states such that the larger the combined Moebius strips appear, the higher the probability of Flow states in both participants. Participants can observe their respective Flow state, as well as one of their interlocutors in real-time. Results show a significant effect of the intervention on social connectedness and perceived closeness as well as on mood. Performing a two-tailed Paired T-test showed significant p-values of .04129022 for involvement and 0.01282582 for happiness for the Somerset House cohort, as well as significance for involvement $p=0.004734$. Performing a two-tailed Paired T-test showed significant p-values of 0.03227 for social connectedness as measured via the IOS survey [Aron et al., 1992]. The control / second cohort (Royal Society, $N=16$) also showed a significant correlation between facilitated participatory engagement and social

connectedness for both the IOS survey $p=0.003573$ and closeness to the other participant $p=0.042421$.

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